

Lecture 24 Divergence Theorem

$$\vec{F} = \langle F_x, F_y, F_z \rangle$$

$$\text{Divergence: } \nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

$$\text{Proposition 1. Curl of the gradient } \nabla \times \nabla f = \vec{0}$$

$$\begin{aligned} \text{Pf. } \nabla \times \nabla f &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \vec{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \vec{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \vec{k} \\ &= \vec{0} \end{aligned}$$

$$\text{Proposition 2. Divergence of the Curl } \nabla \cdot (\nabla \times \vec{F}) = 0$$

$$\begin{aligned} \text{Pf. } \nabla \times \vec{F} &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{i} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \vec{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k}. \\ \nabla \cdot (\nabla \times \vec{F}) &= \frac{\partial}{\partial x} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \\ &= \underbrace{\frac{\partial^2 F_z}{\partial x \partial y}} - \underbrace{\frac{\partial^2 F_y}{\partial x \partial z}} + \underbrace{\frac{\partial^2 F_x}{\partial y \partial z}} - \underbrace{\frac{\partial^2 F_z}{\partial y \partial x}} + \underbrace{\frac{\partial^2 F_y}{\partial z \partial x}} - \underbrace{\frac{\partial^2 F_x}{\partial z \partial y}} \\ &= 0. \end{aligned}$$

Proposition. Product rule for the divergence

$$\nabla \cdot (u \vec{F}) = \nabla u \cdot \vec{F} + u (\nabla \cdot \vec{F})$$

$$(fg)' = f'g + fg'$$

$$\text{Pf. } \nabla \cdot (u \vec{F}) = \frac{\partial (u F_x)}{\partial x} + \frac{\partial (u F_y)}{\partial y} + \frac{\partial (u F_z)}{\partial z}$$

$$\begin{aligned}
&= \underbrace{\frac{\partial u}{\partial x} F_x + u \frac{\partial F_x}{\partial x}}_{\text{div}_x} + \underbrace{\frac{\partial u}{\partial y} F_y + u \frac{\partial F_y}{\partial y}}_{\text{div}_y} + \underbrace{\frac{\partial u}{\partial z} F_z + u \frac{\partial F_z}{\partial z}}_{\text{div}_z} \\
&= \langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \rangle \cdot \langle F_x, F_y, F_z \rangle + u \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \right) \\
&= \nabla u \cdot \vec{F} + u(\nabla \cdot \vec{F})
\end{aligned}$$

Green's Theorem (Circulation form)

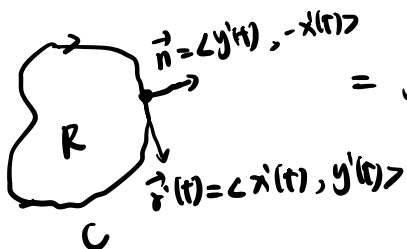
$$\vec{F} = \langle F_x, F_y \rangle \quad \oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} \, dA$$

$$\oint_C F_x dx + F_y dy = \iint_R \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dA$$

Let $\vec{u} = \langle u_x, u_y \rangle = \langle F_y, -F_x \rangle$, i.e., $F_x = -u_y$, $F_y = u_x$

$$\begin{aligned}
\iint_R \underbrace{\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} \right)}_{\nabla \cdot \vec{u}} dA &= \oint_C -u_y dx + u_x dy \\
&= \oint_C -u_y x'(t) dt + u_x y'(t) dt
\end{aligned}$$

$$= \oint_C \langle u_x, u_y \rangle \cdot \langle y'(t), -x'(t) \rangle dt.$$



$$\vec{r}(t) = \langle x(t), y(t) \rangle.$$

$$= \oint_C \vec{u} \cdot \vec{n} \, dt \quad \text{called the flux of } \vec{u} \text{ across } C$$

$$\Rightarrow \text{Green's Theorem (Flux form)} \quad \iint_R (\nabla \cdot \vec{F}) \, dA = \oint_C \vec{F} \cdot \vec{n} \, dt$$

Theorem (Divergence Theorem) Let \$S\$ be a closed orientable surface enclosing a simply connected solid region \$D\$. Suppose \$\vec{F}\$ is a vector field defined and \$C^1\$ in the region \$D\$, then we have,

$$\underbrace{\oint_S \vec{F} \cdot \vec{n} \, ds}_{\text{surface integral}} = \underbrace{\iiint_D \nabla \cdot \vec{F} \, dV}_{\text{triple integral}}.$$

Here \$\vec{n}\$ is the outward normal of \$S\$

Remark: $\oint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds = \iiint_D \nabla \cdot (\nabla \times \vec{F}) \, dv = 0$

Example. Consider the vector field $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$. Let S be the sphere with radius a centered at the origin. Evaluate $\oint_S \vec{F} \cdot \vec{n} \, ds$.

Sol. $\nabla \cdot \vec{F} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3$

By the Divergence Theorem,

$$\oint_S \vec{F} \cdot \vec{n} \, ds = \iiint_D \nabla \cdot \vec{F} \, dv = \iiint_D 3 \, dv = 3 \iiint_D dv = 3 \cdot \frac{4}{3} \pi a^3 = 4\pi a^3.$$

Direct computation:

$$\vec{r} = \langle a \sin \varphi \cos \theta, a \sin \varphi \sin \theta, a \cos \varphi \rangle \quad 0 \leq \varphi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

$$\frac{\partial \vec{r}}{\partial \varphi} \times \frac{\partial \vec{r}}{\partial \theta} = \langle a^2 \sin^2 \varphi \cos \theta, a^2 \sin^2 \varphi \sin \theta, a^2 \sin \varphi \cos \varphi \rangle$$

$$\vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial \varphi} \times \frac{\partial \vec{r}}{\partial \theta} \right) = a^3 \sin^2 \varphi (\cos^2 \theta + \sin^2 \theta) + a^3 \sin \varphi \cos^2 \varphi = a^3 \sin \varphi$$

$$\begin{aligned} \oint_S \vec{F} \cdot \vec{n} \, ds &= \iint_D \vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial \varphi} \times \frac{\partial \vec{r}}{\partial \theta} \right) dA = \int_0^{2\pi} \int_0^\pi a^3 \sin \varphi \, d\varphi \, d\theta \\ &= a^3 2\pi \int_0^\pi \sin \varphi \, d\varphi \\ &= 4\pi a^3. \end{aligned}$$

Example. Let $\vec{F} = x^2\vec{i} + 4xyz\vec{j} + ze^x\vec{k}$, D be the rectangular box defined by $0 \leq x \leq 3$, $0 \leq y \leq 2$, $0 \leq z \leq 1$ and S be the boundary surface of D .

Evaluate $\oint_S \vec{F} \cdot \vec{n} \, ds$.

Sol. $\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(4xyz) + \frac{\partial}{\partial z}(ze^x) = 2x + 4xz + e^x$

By the Divergence Theorem

$$\oint_S \vec{F} \cdot \vec{n} \, ds = \iiint_D \nabla \cdot \vec{F} \, dv = \int_{z=0}^{z=1} \int_{y=0}^{y=2} \int_{x=0}^{x=3} (2x + 4xz + e^x) dx dy dz = 34 + 2e^3$$

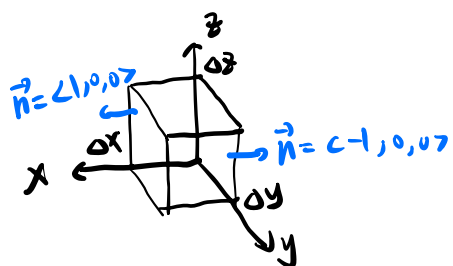
Interpretation of the divergence using mass transport.

$\vec{v}$: velocity field of a fluid $(m s^{-1})$

ρ : its constant density $(g m^{-3})$

$\vec{F} = \rho \vec{v} = \langle F_x, F_y, F_z \rangle$ describes the mass transport of the fluid. $(g m^2 s^{-1})$

$\vec{F}$ multiplying by an area gives the flux $(g s^{-1})$



$$\begin{aligned} & \vec{F}(\Delta x, y, z) \cdot \vec{n} \Delta y \Delta z + \vec{F}(0, y, z) \cdot \vec{n} \Delta y \Delta z \\ &= F_x(\Delta x, y, z) \Delta y \Delta z - F_x(0, y, z) \Delta y \Delta z \\ &= \frac{F_x(\Delta x, y, z) - F_x(0, y, z)}{\Delta x} \Delta V \quad \Delta V = \Delta x \Delta y \Delta z \end{aligned}$$

$$\begin{aligned} & \Delta x \rightarrow 0 \quad \frac{\partial F_x}{\partial x}(0, y, z) \Delta V \\ & \approx \end{aligned}$$

When $\Delta x, \Delta y, \Delta z \rightarrow 0$, net flux out of the cube

$$\approx \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \right) \Big|_{(0,0,0)} \Delta V$$

$$= (\nabla \cdot \vec{F}) \Big|_{(0,0,0)} \Delta V$$

Divide a region D into n small cubes,

$$\text{net flux out of a region } D \approx \sum_{k=1}^n (\nabla \cdot \vec{F})_k \Delta V$$

$$\begin{aligned} & \Delta V \rightarrow 0, \quad n \rightarrow \infty \\ & \approx \iiint_D \nabla \cdot \vec{F} \, dV, \end{aligned}$$

which is also given by $\iint_S \vec{F} \cdot \vec{n} \, dS$.